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The Impact of Economic Integration on Cross-Border Venture Capital Investments: Evidence From the European Union

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Despite major policies surrounding economic integration among countries (e.g., the European Union [EU], North Atlantic Free Trade Area, and Mercosur), the theoretical and empirical research addressing the impact of such policies on various countries' entrepreneurial activities has yet to fully emerge. To address this issue, this paper draws on institutional, economic, and entrepreneurship literatures to examine if two specific economic integration mechanisms, namely market and currency commonality, increase cross-border venture capital flows made by participating nations in the EU. Findings suggest that broad scale economic integration policies do influence the extent of foreign venture capital investments made into other member countries.

Introduction

An enduring, underlying thesis in the fields of business and economics is that entrepreneurial activity is central to long-term economic stability, growth, and development (e.g., Hayek, 1945; Kirzner, 1973, 1979). Recently, the relationship between entrepreneurship and economic growth has seen increased interest at the state and national levels (e.g., Henderson, 2002; Reynolds et al., 2005; van Stel, Carree, & Thurik, 2005). Discussion in this area has centered principally on the idea that governments seeking economic renewal can stimulate economies by reducing financial constraints on the entrepreneur, thus creating increased levels of entrepreneurial activity (Bygrave & Minniti, 2000; Mitchell et al., 2002). Indeed, entrepreneurship has proven to be an essential part of economic growth because of its impact on new job creation, increased trade, and

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heightened levels of competition (Henderson; Minniti, 1999; Mitchell et al.; Reynolds, Hay, & Camp, 1999).

In order to achieve such positive economic effects on a larger scale, a number of national regions have sought to increase economic momentum by creating economically integrated regions across several nations (e.g., the European Union [EU], the North American Free Trade Area, the Southern Common Market, and the Common Market of Eastern and Southern Africa). Economic integration includes such mechanisms as eliminating tariffs, developing common markets, and creating common currencies, which all support market incentives for entrepreneurial strategies (Bosma, Jones, Autio, & Levie, 2008). Generally speaking, economic integration has a positive impact on the overall national economies of the member countries (e.g., Krugman & Obstfeld, 2002) and works to stimulate the reinforcing effects of globalization and entrepreneurship (Jones, 2006). For example, being part of the “Euro zone” makes it possible for the EU members to transfer economic resources from countries with healthy economies to those suffering economic setbacks; such actions work to improve the aggregate economic situation of the integrated area in the long run. Also, such globalization mechanisms tend to vastly increase the integration of financial capital markets among involved countries. Such trends have generally improved the efficiency of financial markets for all participants as capital is now better able to move to areas where it receives the highest reward (Jones).

While economic integration has been utilized extensively throughout the world, it has gained relatively no attention in the field of entrepreneurship, despite a seemingly important relationship. In general, economic integration removes trade barriers and allows goods, services, capital, labor, and technology to move more freely between member countries (e.g., Rose, 2000; Rose & Van Wincoop, 2001). These outcomes, which are empirically and theoretically associated with the integration of economies, tend to promote entrepreneurial activity across regions within nation states (e.g., Bosma et al., 2008; Carree & Thurik, 2003; van Stel et al., 2005). More specifically, integration reduces the uniqueness of institutional structures among countries and increases access to research and educational facilities, sources of financing, and skilled labor pools, which improves the level of entrepreneurial activity (Bartholomew, 1997). Furthermore, an infrastructure that enhances cooperation among entrepreneurs of different countries facilitates problem-solving and cross-border entrepreneurial activity (Casson, 1990). Likewise, since entrepreneurship is typically at the cutting edge of technology, innovation, and new market development, policies restricting trade across markets are generally biased against entrepreneurial activities (Busenitz, Gomez, & Spencer, 2000; Nelson, 1982). Thus, as Kreft and Sobel (2005) argue, entrepreneurship may serve as the “missing link” between the integration of economies across geographic regions and economic growth.

Based on these foundational arguments, this study seeks to take the first step toward understanding how specific government trade policies influence cross-border entrepreneurial activity throughout a multinational region. Specifically, we question whether higher levels of economic integration, through common market and common currency mechanisms, stimulate cross-border venture capital (VC) outflows. VC outflows refer to the aggregated VC investments flowing from traditional private equity/venture firms and corporate venture firms into foreign new ventures, and to the coinvesting by venture firms and corporate venture firms across borders. These investment flows represent primary investments in private ventures and are *not* secondary public market purchases. We utilize VC investment flows because extant literature has shown VC investments to be an important determinant of entrepreneurial activity (Bruining & Wright, 2002; Henderson, 2002; Kreft & Sobel, 2005) and because research dealing with the crossing of national

borders by VC firms has been identified as a “major research gap” in the existing literature (Wright, Pruthi, & Lockett, 2005, p. 135).¹

This paper proceeds as follows. First, building on more general research linking institutional theory, economics, and entrepreneurship, as well as previous studies that examine the factors (e.g., geographic distance, cultural distance, venture stage, VC firm size, and financial relevance) affecting the level of VC activity in foreign markets (Maula & Mäkelä, 2003; Meyer & Shao, 1995), we develop testable hypotheses regarding our research question. Specifically, our research question asks whether the economic integration mechanisms of common market and common currency increase the level of cross-border VC investments in entrepreneurial ventures. Next, we test these hypotheses by following national changes in cross-border VC investing across time at various levels of EU integration. Finally, we conclude the manuscript with a discussion of the key findings, paying particular attention to governmental policy implications.

Hypotheses Development

The link between entrepreneurship and economic growth has long been theoretically recognized (e.g., Schumpeter, 1934). Empirically, studies based on self-employment rates or on the Global Entrepreneurship Monitor database have begun to establish the key role that entrepreneurship plays in national economic growth (e.g., Blanchflower, 2000; Carree, van Stel, Thurik, & Wennekers, 2002; van Stel et al., 2005; Zacharakis, Bygrave, & Shepherd, 2000). Additionally, research has more specifically demonstrated that, among other ways, entrepreneurs affect national economic growth through the introduction of innovations and new products or processes (Acs & Audretsch, 2003), increasing competition (Nickell, 1996), and the creating of knowledge spillovers (Audretsch & Feldman, 1996; Audretsch & Keilbach, 2004).

Despite these studies validating a relationship between entrepreneurial activity and economic growth, it is still rather unclear how nations, through specific governmental policy, might stimulate entrepreneurial activity. One possible option for policy makers is to involve the country in regional economic integration. Economic integration, largely achieved through regional trade agreements (RTAs), is an increasingly common phenomenon occurring across the globe (World Bank, 2005). RTAs are defined as actions taken by governments to liberalize or facilitate trade on a regional basis, oftentimes through the creation of free-trade areas and customs unions (Levy, 2006). Following the theory of comparative advantage among nations (Scharpf, 1996), integration is expected to boost competition, promote growth in certain industries, and positively impact national economies (Krugman & Obstfeld, 2002; Rose, 2000; Rose & Van Wincoop, 2001).

Recent works in the entrepreneurship literature (e.g., Henderson, 2002; Minniti, 1999; Mitchell et al., 2002; Reynolds et al., 1999), have argued that multinational economic integration mechanisms may also promote entrepreneurship in similar ways. In other words, as countries enter greater levels of regional economic integration, entrepreneurs are encouraged, both inside and outside of established firms, to exploit opportunities in the new, broader marketplace (Bygrave & Minniti, 2000). Thus, entrepreneurship may act as

1. While VC investment has been researched extensively and shown to be a major factor in facilitating high growth entrepreneurship in many nations (Bruton, Ahlstrom, & Singh, 2002; Reynolds, Bygrave, Autio, Cox, & Hay, 2002), we acknowledge that VC is limited in many industries and represents a relatively small amount of overall funding in many areas of business development. We thank an anonymous reviewer for bringing this point to our attention.

a mediator or catalyst between integration policies and economic growth (Minniti). We contribute to this literature by suggesting that the adoption of a common market and a common currency are two key integration mechanisms that are positively associated with the level of cross-border VC investment occurring between involved countries and, therefore, to the entrepreneurial development within those associated countries. This is important because there is evidence that the behavior of venture capitalists is central to the increased levels of entrepreneurial activity occurring within and between nations by encouraging the flows of capital and ideas (Barry, 1994; Bygrave & Timmons, 1999; Kreft & Sobel, 2005; Manigart, 1994).

Venture capitalists have shown increasing interest in crossing borders to gain access to investment opportunities, as shown by the growing level of activity internationally. According to Gompers (2005), for example, half of new VC funds originate from nondomestic sources in Asia and Europe, while less than 10% of VC funds are nondomestic in the United States. Further, the share of VC inflows from nondomestic sources more than doubled in Europe from 1988 to 2003, while outflows almost quadrupled during the same time frame (Wright et al., 2005). Of course, the investment activity of venture capitalists is contingent on a number of institutional factors such as the development of stock markets and legal infrastructures within the target countries (Wright et al., 1993, 2005). This is why the level of cross-border VC investments is importantly linked to broad-sweeping changes in the institutional environment that occur because of economic integration. Hence, we develop an institutional perspective on VC markets to provide a greater understanding of the relationship between integration mechanisms and VC investment flows.

The institutional framework of a society is comprised of the fundamental legal, social, and political practices and rules that establish the basis for individual and organizational operations (Manalova, Eunni, & Gyoshev, 2008). Institutional factors such as a well-developed stock market system, which allows venture capitalists the ability to exit via initial public offerings (IPOs) (Black & Gilson, 1998); supportive regulatory frameworks (Jeng & Wells, 2000; Megginson, 2004); temperate legal environments, particularly regarding bankruptcy (Armour & Cummings, 2006); as well as social and cultural factors (Wright et al., 2005), all serve to stimulate entrepreneurship and the demand for VC (Bruton, Fried, & Manigart, 2005; Hwang & Powell, 2005). Integration directly impacts these institutional factors by creating increased similarity among the participating countries, and therefore influences the VC investment flows.

This study argues that by entering into higher levels of economic integration, countries are likely to stimulate their cross-border entrepreneurial activity since such activity is directly associated with cross-border VC investment flows to and from other member nations (Bottazzi, Da Rin, & Hellman, 2004; Kreft & Sobel, 2005). As Fligstein and Sweet (2002) argue, the first stage of the EU, the Treaty of Rome, was specifically designed to promote economic exchange across national borders. Indeed, these authors further identified a self-reinforcing cycle of reduced barriers to cross-border economic activity and increasing integration. Specifically, Fligstein and Sweet state “that the process through which barriers to cross-border economic activity within Europe are removed . . . and the process through which common, supranational public policies are made and enforced . . . are not distinct processes” (pp. 1209–1210), but are endogenous to one another. Following these arguments, as well as those made in the subsequent paragraphs, we expect countries to exhibit increased cross-border VC investments to other member countries as they join the common market and adopt the common currency of the union. But while conceptually distinct, these two mechanisms generally function similarly to increase the resemblance in regulatory and legal institutional structures between countries, one through market systems and the other through financial infrastructures.

Building on the institutional framework described above, two perspectives lend support to our arguments. First, based on the premise that organizational managers are both growth-seeking and risk-minimizing and foreign market uncertainty is the major deterrent for foreign market entry (Johanson & Vahlne, 1977), the Uppsala Internationalization Model argues that firms will initially develop domestic markets and then incrementally expand operations abroad (Leonidou & Katsikeas, 1996). We can apply the same logic to VC firms and argue that foreign market uncertainty leads VC investors, who already have a high degree of uncertainty because of their involvement in new and unproven ventures, to be even more cautious. In other words, venture capitalists are more likely to invest in ventures with which they have more information and experience regarding key components of the market (Higgins & Gulati, 2006; Hymer, 1960).

A second related perspective that lends itself to improving our understanding of the relationship between integration mechanisms and VC investment outflows, is one based on theories of cognition and social information processing (cf. Busenitz & Lau, 1996). Recent studies in this area have examined the cognitive aspects of entrepreneurial thinking, where “entrepreneurial cognitions are the knowledge structures that people use to make assessments, judgments, or decisions involving opportunity evaluation, venture creation, and growth” (Mitchell et al., 2002, p. 97). Related to decision making by venture capitalists, entrepreneurial cognition research suggests that decision makers use simplified mental models to more easily piece together important pieces of information in order to help them identify and develop new opportunities. In other words, as venture capitalists become more familiar with the information surrounding a decision, the information processing becomes more automated. In contrast, as the information surrounding a decision becomes more complex and unfamiliar, venture capitalists will be forced to operate outside of their normal procedures, which reduces confidence in making a particular decision (Zacharakis & Shepherd, 2001). Overall, it may be argued that lowering cognitive transaction costs for entrepreneurs can stimulate more entrepreneurial activity (Mitchell, 2003; Mitchell, Smith, Seawright, & Morse, 2000; Mitchell et al.) or in this case, stimulate VC investing in new and unproven markets.

Building on these related perspectives, related research has utilized the transaction cost perspective of multinational enterprises to argue that contextual uncertainty, which entails bounded rationality and arises from institutional changes in national environments, influences international entry mode decisions (Yiu & Makino, 2002). Therefore, among other benefits, adoption of an integrated market limits uncertainty and risk associated with markets and institutions by reducing institutional differences among participating countries. Consequently, we expect integration to reduce market uncertainty and reduce market risk to increase VC flows between member states. Likewise, reductions in the complexity associated with VC transactions should be associated with increased levels of investment because of the reduction in costs associated with financial exchange transactions and the risks associated with exchange rate fluctuations and inflation levels. In fact, an efficient financial structure is necessary for the proper mobilization of domestic and foreign savings toward investment opportunities, while an inefficient financial sector constrains economic growth and entrepreneurship by undermining investor confidence (Bencivenga & Smith, 1991).

In summary, when considering cross-border VC investments, the greater the similarity between two countries’ institutional systems and structure, the more likely a venture capitalist will recognize and take advantage of opportunities existing across national borders. Thus, the lowering of trade barriers through the adoption of a common currency would stimulate cross-border VC outflows by reducing the number of transaction-based

and risk-related barriers that exist for investors. Given these arguments, we formally state the following hypotheses:

Hypothesis 1: A country will have higher levels of cross-border venture capital investment outflows after the country adopts the common market of the regional economic community.

Hypothesis 2: A country will have higher levels of cross-border venture capital investment outflows after the country adopts the common currency of the regional economic community.

Context and Data

The EU

The EU is an important and timely illustration of a region attempting to integrate economically. The EU is an emerging economic and political superpower of the same order as the United States and Japan, and it is reshaping the direction of international trade and foreign direct investment (FDI). Indeed, the EU has become the fourth addition to the old international trade triad and, as of 2007, has emerged as the world's largest economy (CIA World Factbook, 2007).

According to international economic theories that address different levels of regional economic integration, the current status of the EU fits the category of an economic union, which by definition involves the free flow of products and factors of production between member countries and the adoption of both a common external trade policy and a common currency. However, the EU is not a "perfect" economic union since (1) not all members have adopted the common currency, and (2) differences in tax rates across countries still remain (Hill, 2002). Further, if examined over time, the EU and its member countries continue to evolve through several transitions regarding economic treaties and trading arrangements. As such, the EU serves as a highly visible and relevant context within which to test the previously developed hypotheses.²

Sample and Data

The data used in this study were collected from 24 countries for the period between 1985 and 2002. These 24 countries consist of the EU-15, six recently admitted members of the EU-25, and three countries that are not official members of the EU but represent large economies located in Europe (i.e., Iceland, Norway, and Switzerland).³ The 18-year

2. While the EU originally started with six countries, it has increased its membership to 27 countries. The first six members of the European Steel and Coal Community included Belgium, the Federal Republic of [West] Germany, France, Italy, Luxembourg, and The Netherlands. The European Economic Community was later joined by Denmark, the United Kingdom, and Ireland in 1973. In 1981, Greece joined the Community. Spain and Portugal followed in 1986. In 1995, Austria, Finland, and Sweden joined the EU. Throughout the 1990s, the EU received 12 applications for membership consideration: two applicants from the Mediterranean countries (Cyprus and Malta [1990]), nine applicants from former Soviet republics (Hungary and Poland [1994] and Slovakia et al. [1995] and the Czech Republic [1996]), and one applicant from a former Yugoslavian republic (Slovenia [1996]). In 2004, 10 of the above listed countries joined the EU: Cyprus, Czech Republic, Estonia, Hungary, Latvia, Lithuania, Malta, Poland, Slovakia, and Slovenia. Bulgaria and Romania did not make the final round in 2004, but were added in January of 2007.

3. Cyprus, Latvia, Lithuania, Malta, Slovakia, Bulgaria, Croatia, Yugoslavia/Macedonia, or Turkey were not included due to lack of detailed data regarding VC flows to other countries.

period is pivotal for the empirical context of this study since it represents the period during which most major changes to the EU took place whether in terms of both the level of integration and the number of members. Indeed, this time frame captures the effect of each of these changes, and as a result, serves the purpose of the study. Additionally, this time period predates the emergence of the treaty establishing a Constitution for Europe (TCE), which intended to dispense with the previously overlapping treaties that comprised the current constitution.⁴ Finally, the time period allows 3 years for the effects of the adoption of the Euro on January 1, 1999 to occur. The original sample size is 408 country-years.

Dependent Variable

We collected information regarding VC investments from Securities Data Corporation (SDC) Platinum, a comprehensive database developed and maintained by Thompson Financial Corporation.⁵ The international data on VC investments in SDC Platinum are gathered by Venture Economics. Venture Economics collects data on VC investments in Europe, for inclusion in the SDC Platinum database, by soliciting information from VC associations such as member firms of the British Venture Capital Association (BVCA) and the European Venture Capital and Private Equity Association (EVCA). Hence, the database contains only those transactions that meet criteria considered “standard industry practices” or “standard venture capital” by the private equity industry in Europe. Standard VC investments include those made by traditional private equity/venture firms, corporate venture firms, and coinvesting by venture firms and corporate venture firms, but do not include angel investing by family or friends.⁶ Additionally, VC investment includes primary investments in preseed and seed-staged organizations through later stage, mezzanine funding, and bridge loans. The central characteristic of the investment is that it *not* be a secondary public market purchase. Finally, the BVCA and EVCA guidelines are quite similar and congruent with U.S. industry practices and they represent the vast majority of private equity and VC providers for the United Kingdom and continental Europe, respectively. Thus, SDC Platinum requires no minimum or maximum amount of investment to be included in the database, only that the investment *not* be a public market purchase.

We operationalized the dependent variable as EU cross-border VC investment outflows by taking the ratio of the VC invested by each country in other EU countries to the total VC invested by that country. Table 1 presents more detail on the dependent variable by listing VC investment characteristics by the countries used in this study. Values are averaged across the sample time frame from 1985 to 2002. From this table, we observe significant variation in the size and number of VC investments across European countries that must be accounted for in our statistical model. For example, VC investments to other

4. The TCE was initially signed in 2004 by all representatives of the EU, but ended up being rejected in referenda. Had the treaty been accepted, it would have simplified and replaced the existing set of overlapping treaties and thus significantly reduced the variance associated with the EU members regarding legal and economic policy. Given that all members initially signed the treaty, there is reason to believe that the institutional differences in the countries had already started to dissolve greatly. Limiting the study to 2002 reduces the impact the impending treaty might have had on the involved countries.

5. SDC Platinum is a collection of databases with information on financial transactions. In addition to VC transactions, the database includes data on new issues, mergers and acquisitions, alliances, and corporate governance.

6. Informal investments, which include support from friends and family, have been found to be a major source of funding for early-stage entrepreneurial ventures (Bygrave, Hay, & Reynolds, 2003; European Commission, 2002; Mason & Harrison, 2000; Reynolds et al., 2002). In 2007, informal funding to new firms in 37 countries was estimated at \$298 billion, compared to \$59 billion from formal VC investments (Reynolds et al.).

Table 1

Venture Capital Investment Characteristics by Country*

Country	Average [†] investment (\$)	Average minimum investment (\$)	Average maximum investment (\$)	Count of investments
Austria	2.49	.50	7.78	349
Belgium	5.63	.96	11.47	976
Czech Republic	3.06	.43	2.55	151
Denmark	3.13	.42	13.03	1,025
Estonia	2.02	1.78	2.02	15
Finland	5.63	.90	7.95	1,570
France	8.20	3.85	16.39	6,250
Germany	5.30	3.23	13.10	3,355
Greece	3.10	1.09	4.55	65
Hungary	2.08	.81	4.53	105
Iceland	1.01	.03	9.53	25
Ireland	4.82	.32	6.18	627
Italy	4.92	2.06	21.44	582
Luxembourg	13.03	1.63	17.02	327
The Netherlands	10.71	.81	27.21	2,109
Norway	4.76	.92	7.09	596
Poland	6.33	1.29	9.96	424
Portugal	13.33	.71	33.70	381
Romania	6.16	1.37	4.97	45
Slovenia	3.03	1.25	3.03	18
Spain	15.27	1.73	39.97	649
Sweden	14.09	5.79	23.84	1,704
Switzerland	6.25	1.44	8.42	1,112
United Kingdom	23.33	2.08	29.19	13,755
Overall average	6.99	1.47	13.54	1,509

* Values are in U.S.\$ millions.

† All averages represent the average of yearly values 1985–2002.

EU countries across the time period of the study range from an average minimum of \$1.47 million to an average maximum of \$13.54 million. Finally, the average investment made by a VC firm is \$6.99 million between 1985 and 2002.

Figure 1 visually illustrates that the overall changes in total VC expenditures in the EU tend to mirror those in the United States and have recently begun to closely rival those in the United States. Understanding this similarity is important to establish that VC investing patterns and behaviors are not U.S. specific, but that international VC is an increasingly important phenomenon occurring worldwide (Wright et al., 2005). Figure 2 displays the targets of VC outflows specific to our data; this figure shows a decrease in European VC outflows compared to the United States, as well as a reduction of VC flows that remain within a specific EU country over the last few years of the study. Figure 2 also shows that VC outflows in countries outside of EU and United States increased slightly. Finally, the amount of VC outflows from EU countries to other EU countries remained fairly stable over time, in spite of some cyclical behavior after 2000. Given the significant reduction in overall VC investment between 2000 and 2002, investments flowing to other EU countries during this period represent an increased percentage of overall VC investments.

Figure 1

Comparison of Venture Capital Investments as a Percentage (%) of Gross Domestic Product Between EU and United States. Source: Eurostat (<http://ec.europa.eu/eurostat/>)

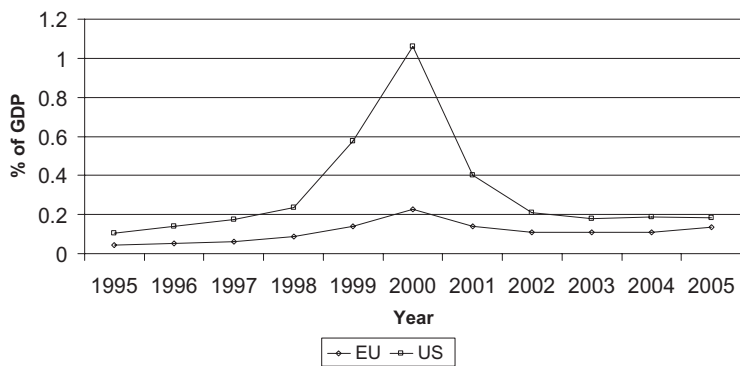
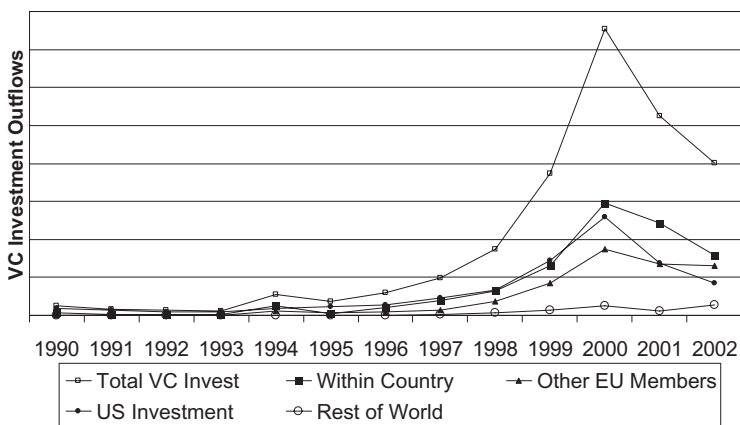


Figure 2

Comparison of Venture Capital (VC) Investment Outflows Specific to Study Data. Source: SDC Platinum



Although preliminary, these patterns suggest that more recent VC investments within the EU have been targeting opportunities outside of the United States and their own borders. Rather, an increased percentage of VC investment outflows have been toward other EU countries and foreign countries other than the United States, which has historically held a large portion of VC investments from foreign countries. Such trends support the trends noted by Wright et al. (2005) and demonstrate the growing importance of VC investment inflows and outflows to both policy makers and researchers.

Independent Variables

The information for the independent variables used in this study was collected from data compiled by the Statistical Office of the European Communities (Eurostat).⁷

Adoption of a Common Market. This variable indicates whether a country is part of the single market for goods, services, and factors of production among the EU community. The European Economic Community (later becoming the European Union) was founded in 1957 when member states signed the Treaty of Rome, creating a “common market” for trade. The resulting single market allowed goods, people, and services to move freely throughout the EU. Specifically, it indicates whether a country has removed the tariff and nontariff barriers to trade with the rest of the EU members. It is coded as a dichotomous variable where (1) is assigned for countries that share the single market with the rest of the EU members and (0) for countries that are not part of this single market of goods and services.

Adoption of a Common Currency. In 1991, the member states approved the Maastricht Treaty to establish the European Monetary Union, which made the Euro the EU’s common currency. The Euro was adopted to promote closer economic and monetary cooperation, so that the common market could develop and flourish further (Leblond, 2004). Accompanying the adoption of the Euro, European Central Bank was established with the mandate to maintain price stability (Baldwin, 1991; Salvatore, 2002). This variable indicates whether a country has agreed to adopt the Euro as its formal currency, since not all EU members belong to the Euro zone (e.g., the United Kingdom and Denmark). This variable is coded as a dichotomous variable with (1) for nations that adopted the Euro at a specific year and (0) for the nations that did not adopt the Euro.

Control Variables

The study employs several variables to account for other explanations of the pattern of VC investments among organizations of EU member countries. These variables are country level variables that control for the overall differences between countries in terms of economic growth, transition status, and overall business environment. We obtained information for these variables from the World Bank, the World Trade Organization, the International Monetary Fund, the CIA World Factbook, and Eurostat.

Gross Domestic Product (GDP) Growth. Accounting for the relative wealth of a country is of major concern in this study. Trends in international trade suggest that more developed countries are more likely to invest or trade with other more developed or wealthier countries (International Monetary Fund, Statistics Department, 2004). Accordingly, we used the annual percentage growth of the GDP of each of the countries included in this study over the period 1985–2002 as a control variable. This variable measures the overall economic improvement and wealth of each of these nations and allows us to control for the effect of economic wealth of any two countries on the pattern of cross-border VC outflows.

7. Eurostat, established in 1953, supplies statistics to the European Commission and other institutions with the goal of defining, analyzing, and implementing macroeconomic policies. Data used in this study are available to the public at <http://ec.europa.eu/eurostat/>.

Inflation Levels. Macroeconomic theories propose that high levels of inflation encourage borrowing since borrowers calculate the real interest rate by subtracting the inflation rate from the nominal interest rate. This borrowing premium suggests that countries with higher inflation levels might show higher levels of private borrowing by VCs within the country. This borrowing could be potentially used for foreign investment purposes. To account for this possibility, we use the inflation rate of each country as a control variable.

FDI and Balance of Payment (BOP). FDI and BOP figures of a nation indicate the amount of international activities of each nation. We use these two variables as control variables to isolate the effect of the intensity of a country's involvement in international trade and investments. These two activities suggest that an organization within a country that is highly involved with international markets is more likely to seek out investment opportunities outside of the focal country. Several studies have empirically demonstrated the importance of FDI to economic growth (e.g., Bengoa & Sanchez-Robles, 2003; Borensztein, De Gregorio, & Lee, 1998).

Transitioning Countries. Finally, we include a control variable to identify whether or not the country is a politically transitioning country. Specifically, we are concerned about the social and cultural institutional differences that may exist between former communistic countries and western countries. As more Eastern European countries join the EU, there is a possibility that communistic political ties may influence the behaviors of VC investors because political regimes have an effect on the forms and patterns of investment strategies exercised by individuals and organizations (Hyder & Abraha, 2006; Jensen, 2003). To control for this possibility, we include a dichotomous control variable in the models where (1) indicates that the country was formally a communist country, and (0) indicates that the country was not formally under a communist political regime.

Founding Member. This measure indicates whether a specific country is one of the original six countries of the European Coal and Steel Community that initially established the EU. We created a dichotomous variable to control for this designation because EU founding members gained full access to the economic integration by default, while joining members, in order to be members of the European monetary system, had to undergo and implement macroeconomic reforms pertaining to foreign debt, unemployment, and the banking sector. Thus, the adoption of the common currency and the integration to the European common market for *joining* members typically occurs after these countries make several macroeconomic reforms in the banking, financial, and enterprise sectors (Scharpf, 1996). The measure is coded as a value of (1) for France, Italy, Luxembourg, Germany, The Netherlands, and Belgium and a value of (0) for the remaining 18 countries included in this study.

Analyses and Results

In models that violate the assumption of independence of errors, care must be taken to choose appropriate modeling techniques that reduce this impact. When data is collected on the same set of subjects over a period of time, observations on the same subject are expected to be correlated (Johnson & Wichern, 2002). Thus, observations with temporal dependence violate the assumption of independence of errors. Correlated errors result in spurious results due to inflated error terms (Beck & Katz, 1995). In this paper, given that we conducted our study across the same 24 countries over a period of 18 years, the data

is characterized by repeated measures both in terms of year and in terms of countries. Consequently, observations collected over one country are expected to exhibit time series correlations and observations collected for a specific year are expected to exhibit cross-sectional correlations.

With observational studies, as opposed to classically designed experiments with balanced and complete data, it is rarely possible to achieve complete separation of between- and within-subject variables. Complete separation of between- and within-subject variables is particularly important in our case because we are concerned with patterns across multiple time periods (within-subjects) and across countries (between-subjects). Therefore, due to multiple courses of correlation and nonconstant variability in error terms, we employ a linear mixed model designed to establish the effect of economic integration (i.e., common market and common currency) on changes in cross-border VC outflows over time. Linear mixed models expand the general linear model so that data are permitted to exhibit correlated and nonconstant variability in error terms. In other words, the mixed linear model provides the flexibility of modeling not only the means of the data but also their variances and covariances.

The mixed linear model we use in our analyses is a version of a multilevel linear model (Goldstein, 1987) or hierarchical linear model (Bryk & Raudenbush, 1992). We use a general linear mixed model in order to deal with violations of independence, which as mentioned earlier, tend to inflate standard errors and make finding significance more difficult (Krull & MacKinnon, 2001). Specifically, we employ an autoregressive design capable of controlling for the nonindependence of error terms. We chose an autoregressive design rather than a standard multilevel model because we expected nonconstant variability (in addition to nonindependence) in error terms that are caused by cross-sectional and times-series correlations.

The general form for a mixed model is:

$$Y = X\beta + Zu + e$$

where $X\beta$ denotes fixed effects, Z is a fixed design matrix, u is a random unobservable vector of random effects with $E(u) = 0$ and $Cov(u) = G$, and e denotes the random unobservable vector of errors, with $E(e) = 0$ and $Cov(e) = R$.

The linear mixed model used in this study assumes three types of correlations. The first type of correlation is the time-series correlation based on common country effect, where observations in a country share a vector $country_i$. The second type of correlation is the cross-sectional correlations between countries, where countries in the same year share a vector $year_j$. The third type of correlation is the additional time series correlation. Since we expect the correlations to be positive and decay over time, we used the (AR [1]) covariance structure to provide a parsimonious covariance structure.

The general form for the mixed model of this study is:

$$Y_{ij} = \beta_0 + \beta_1 X_{ij} + country_i + year_j + \varepsilon_{ij}$$

where $\beta_0 + \beta_1 X_{ij}$ denotes the fixed effects, $country_i$ induces correlation between years, and $year_j$ induces correlation between countries.

Table 2 contains the correlations between variables while Table 3 contains the results of the mixed models. The first model in Table 3 contains only the control variables utilized in the analysis; here, only FDI showed a significant impact on VC investment outflows ($p < .05$). The Common Market Model (the second model) shows acceptable goodness of fit ($-2\log L = -126.4$) and tests for the effect of joining the EU

Table 3

Mixed Models Results

	Control variables β (SE)	Common market β (SE)	Common currency β (SE)	All variables β (SE)
GDP change	-.001 (.0007)	-.0008 (.0007)	-.0008 (.0007)	-.0007 (.0007)
BOP	-.0008 (.002)	-.001 (.002)	-.00001 (.002)	-.0002 (.002)
FDI	.001 (.0005)**	-.001 (.0005)**	-.0008 (.0005)	-.0007 (.0005)
Inflation	-.0001 (.0001)	-.0001 (.0001)	-.0001 (.0001)	-.0001 (.0001)
Political regime	-.061 (.066)	-.031 (.069)	-.051 (.065)	-.030 (.069)
Founder or joiner	.027 (.057)	-.015 (.062)	.012 (.057)	-.016 (.061)
Common market		.097 (.033)***		.066 (.034)*
Common currency			.192 (.046)***	.165 (.048)***
Model diagnostics				
-2log L	-73.193***	-126.404***	-135.353***	-133.657***
AR(1) diagonal	.056 (.005)***	.055 (.005)***	.053 (.005)***	.052 (.004)***
AR1 rho	.455 (.042)***	.479 (.043)***	.497 (.043)***	.477 (.043)***

* $p < .10$, ** $p < .05$, *** $p < .001$.

GDP, gross domestic product; BOP, balance of payment; FCI, foreign direct investment.

on the percent of cross-border EU investment. The results of the second model support hypothesis 1 ($\beta = .097$, $p < .001$) and suggest that participating in the common EU market significantly increases the amount of VC investment that flows to other EU countries. Similarly, the Common Currency Model (the third model) lends support for hypothesis 2 and represented the data best (2log L = -135.35); the results found here indicate that the adoption of the Euro results in increased cross-border EU VC outflows ($\beta = .192$, $p < .001$).

In order to examine the combined effects of the common market and common currency on VC investments to other EU countries, we constructed a full model with both effects. The final model in Table 3 finds that the adoption of the Euro had the most significant relationship with the percentage of cross-border investments ($\beta = .165$, $p < .001$). Also, joining the common market was moderately significant in the All Variables Model ($\beta = .066$, $p < .10$). Therefore, we find that both economic integration mechanisms result in increased VC investment from a particular EU country to other EU countries; this lends support for both hypotheses. However, adoption of the common currency appears to be the most significant mechanism for stimulating cross-border investment to other EU countries.

In all four models, the AR(1) diagonal and AR(1) rho statistics were highly significant, which supports our assumptions regarding the covariance structure for the residuals. We expected that repeated observations on a subject would be positively correlated and found the AR(1) covariance structure produced parsimonious estimates with acceptable precision. The basic implication is that time trends were estimated efficiently. It appears that estimates of the effects of country level variables across time, including the adoption of the Common Market and Euro, are produced effectively with the AR(1) covariance structure. This suggests that there is fairly rapid decay, since the farther apart the measurements, the lower their correlations.

Discussion and Implications

This exploratory study, which found support for both hypotheses, suggests that changes associated with economic integration at the country level have a direct, positive impact on cross-border VC investment flows within the integrated region; such flows would tend to enhance technology diffusion and facilitate networking that would eventually heighten overall entrepreneurial activity of the individual country and the entire region (Borensztein et al., 1998). Therefore, the research offered in this article lends support for our research question and directly addresses an identified gap in the literature regarding the role institutional environments play on international VC markets (Wright et al., 2005). In this sense, the study extends both economic and international entrepreneurship research.

While we find that both the adoption of a common market and a common currency increase the level of cross-border VC investment outflows, the limitations of this study should be acknowledged. First, it is worth noting that our study is limited to a specific region and a specific RTA. Clearly, the EU represents a unique group of countries in terms of size and proximity to one another. Therefore, the same effects of adopting these two integration mechanisms may not be realized when applied to other national regions or RTAs. Second, a general assumption throughout this manuscript has been that there is a relationship between the VC investment flows and entrepreneurial activity, and ultimately, economic growth. While previous research has demonstrated these linkages, our study does not explicitly test these latter relationships and we can, therefore, only speculate as to these final outcomes. Third, venture capitalists make decisions based on a number of factors, many of which we do not discuss herein or include in our analysis. This study is limited, therefore, in that we only examine aggregate data at the country level and do not consider the numerous other factors that may influence VC investing patterns. Finally, we do not examine the country inflows of entrepreneurial activity. Kreft and Sobel (2005, p. 603), in their study of VC flows across state borders in the United States, demonstrate that VC investment funds flow into states with “already well-established entrepreneurial activity.” Although one cannot draw exact parallels between the Kreft and Sobel study and the study outlined in this manuscript, VC inflows are an area in need of further examination.

While these limitations should certainly be taken into account in assessing the representativeness of the findings, this study does appear to provide some important implications. Most obvious, of course, is the implication for national leaders seeking to develop policies that promote entrepreneurial activity, which, in turn, stimulates economic growth (e.g., Henderson, 2002; Reynolds et al., 1999; Zacharakis et al., 2000). As noted previously, we have only empirically investigated the economic integration activities of the EU. Given this limitation, the study suggests national leaders, particularly of smaller, less developed nation-states, should note the potential benefit of economic integration policies, especially the adoption of common currencies, on VC investment flows. The positive implications of economic integration become imperative as these multilateral agreements provide access to global markets, instead of isolation, for organizations in these countries. Indeed, the results of this study might serve as a meaningful argument against those stakeholders that would lobby against integrating policies. For instance, creating more global markets would force many existing and often large and powerful firms to deal with increased levels of competition due to trade liberalization and an increased number of imports (Jones, 2006).

As previously noted, the findings of this study show that involvement in economic integration can substantially increase the flows of VC investments across national borders. However, our data also seems to suggest that after a country adopts a common market or

a common currency, a larger percentage of its VC investment flows go back into the economically integrated union, rather than to outside countries (e.g., China, Japan, and the United States). As shown in Figure 2, it is likely that overall VC investment outflows from EU nations may be decreasing with regards to the United States, but proportionately increasing with regards to other EU nations. Thus, the intuition may be that any reduction in inhibiting factors that might keep entrepreneurs from entering the associated country or seeking outside (and larger) markets contributes to overall economic stimulation. For although we examine investment activity at the country level, national entrepreneurial activity is simply “the net result of individual decisions to pursue entrepreneurial initiatives” (Reynolds et al., 2005, p. 208). These findings are especially important given reports by the European Commission in 1994, 1998, and 2001 that encourage European governments and institutions to create appropriate environments for developing VC as a way to foster entrepreneurial outcomes such as job creation and international competitiveness (Bottazzi & Da Rin, 2002).

In a related study, De Backer and Sleuwaegen (2003) demonstrate that FDI tends to “crowd out” domestic entrepreneurship. Although we found significant, positive effects of economic integration on cross-border VC investment while controlling for FDI, one might argue that new member nations of an integrated economy might see similar effects regarding the decline of domestic new ventures. In other words, it might be that VC investment outflows may displace some of the entrepreneurial activity from the focal nation to other members of the union. However, De Backer and Sleuwaegen (p. 67) also suggest that the decline in domestic entrepreneurship may be “moderated or even reversed in the long run due to the positive effects of FDI on domestic entrepreneurship as a result of learning, demonstration, networking and linkage effects between foreign and domestic firms.” This moderation or reversal is likely also true for the outflows of VC investments and solidifies its use in this study. Following this line of thinking, future work in this area might more specifically analyze the patterns of VC inflows and outflows on a country by country basis in order to determine what institutional factors contribute to the dissemination of VC. In addition to legal, regulatory, and financial institutional factors, one might consider what similarities or differences in culture, language, religion, or other socio-cultural factors better explain some exchanges by these countries. In other words, once these new markets opened up to the entrepreneurs in a given country, where do they go first or most extensively?

In a related article, Bygrave and Minniti (2000) identified the phenomenon of reinforcing entrepreneurial activity, which implies that as entrepreneurs are more prevalent in a region, more entrepreneurial activity will take place. Based on our findings, we suggest that the same phenomenon may exist for entrepreneurial activity at a multi-country level and across national borders. It seems likely that as people witness others developing new products, services, or firms; investing in firms outside of the focal country; or entering new country markets, they will be more receptive to such behaviors. This argument contributes to the ongoing debate in the entrepreneurship literature about whether there are national differences, such as culture, that significantly impact the level and nature of entrepreneurship (Dodd, 2002; Shane, 1992; Thomas & Mueller, 2000). Recently, efforts regarding cognitive processes have suggested that although there are generic cognitive scripts, the values that shape these scripts may differ greatly according to cultural differences (Busenitz & Lau, 1996; Mitchell et al., 2000). Future research in this area might incorporate the effects of integration and RTAs, so as to note the changes in attitudes and cognitive processes regarding entrepreneurial activity within a country. In other words, to what extent does involvement in an integrated economy minimize the national differences that otherwise differentiate the levels and nature of entrepreneurship between countries?

Conclusion

This study suggests that when countries' economies become more integrated (i.e., as they adopt common markets and currencies), an increase in the amount of cross-border VC investment is likely to follow. As noted by Minniti (1999) and Kreft and Sobel (2005), entrepreneurship may serve as the “catalyst” or “missing link” between the integration of economies and economic growth. Thus, because of the linkages between economic integration and economic growth, we can surmise that entrepreneurial activity, as it relates to VC investment activity, is a key piece of the process. This linkage is an important one, particularly for governmental policy makers who may be considering the pros and cons of economic integration with other nations.

REFERENCES

- Acs, Z.J. & Audretsch, D.B. (2003). Innovation and technological change. In Z.J. Acs & D.B. Audretsch (Eds.), *Handbook of entrepreneurship research* (pp. 55–79). Boston: Kluwer Academic Publishers.
- Armour, J. & Cummings, D. (2006). The legislative road to Silicon Valley. *Oxford Economic Papers*, 58(4), 596–635.
- Audretsch, D.B. & Feldman, M. (1996). R&D spillovers and the geography of innovation and production. *American Economic Review*, 86, 630–640.
- Audretsch, D.B. & Keilbach, M. (2004). Entrepreneurship capital and economic performance. *Regional Studies*, 38, 949–959.
- Baldwin, R.E. (1991). On the microeconomics of the European Monetary Union. *European Economy*, Special Edition No. 1: The Economics of EMU: Background Studies for “One Market One Money,” 44, 21–35.
- Barry, C. (1994). New directions on research in venture capital finance. *Financial Management*, 23(3), 3–15.
- Bartholomew, S. (1997). National systems of biotechnology innovation: Complex interdependence in the global system. *Journal of Business Studies*, 28, 241–266.
- Beck, N. & Katz, J.N. (1995). What to do (and not to do) with time-series cross-section data. *The American Political Science Review*, 89(3), 634–647.
- Bencivenga, V.R. & Smith, B.D. (1991). Financial intermediation and endogenous growth. *Review of Economic Studies*, 58, 195–209.
- Bengoa, M. & Sanchez-Robles, B. (2003). Foreign direct investment, economic freedom and growth: New evidence from Latin America. *European Journal of Political Economy*, 19, 529–545.
- Black, B. & Gilson, S. (1998). Venture capital and the structure of capital markets: Banks versus stock markets. *Journal of Financial Economics*, 47, 243–277.
- Blanchflower, D.G. (2000). Self-employment in OECD countries. *Labour Economics*, 7, 471–505.
- Borensztein, E., De Gregorio, J., & Lee, J.W. (1998). How does foreign direct investment affect economic growth? *Journal of International Economics*, 16, 215–241.
- Bosma, N., Jones, K., Autio, E., & Levie, J. (2008). *Global entrepreneurship monitor: 2007 executive report*. Kansas City, MO: Kauffman Center for Entrepreneurial Leadership.

- Bottazzi, L. & Da Rin, M. (2002). Venture capital in Europe and the financing of innovative companies. *Economic Policy*, 17(34), 229–270.
- Bottazzi, L., Da Rin, M., & Hellman, T. (2004). The changing face of the European venture capital industry: Facts and analysis. *Journal of Private Equity*, 7(2), 26–53.
- Bruining, H. & Wright, M. (2002). Entrepreneurial orientation in management buyouts and the contribution of venture capital. *Venture Capital*, 4(2), 147–168.
- Bruton, G.D., Ahlstrom, D., & Singh, K. (2002). Venture capital in Singapore: Its unique position in Asia, and its future. *Venture Capital: An International Journal of Entrepreneurial Finance*, 4, 197–218.
- Bruton, G.D., Fried, V.H., & Manigart, S. (2005). Institutional influences on the worldwide expansion of venture capital. *Entrepreneurship Theory and Practice*, 29(6), 737–760.
- Bryk, A.S. & Raudenbush, S.W. (1992). *Hierarchical linear models: Applications and data analysis methods*. Newberry Park, CA: Sage Publications.
- Busenitz, L. & Lau, C. (1996). A cross-cultural cognitive model of new venture creation. *Entrepreneurship Theory and Practice*, 20(4), 25–40.
- Busenitz, L.W., Gomez, C., & Spencer, J.W. (2000). Country institutional profiles: Unlocking entrepreneurial phenomena. *Academy of Management Journal*, 43(5), 994–1003.
- Bygrave, W. & Minniti, M. (2000). The social dynamics of entrepreneurship. *Entrepreneurship Theory and Practice*, 24, 25–36.
- Bygrave, W.D., Hay, M., & Reynolds, P.D. (2003). Executive forum: A study of informal investing in 29 nations composing the Global Entrepreneurship Monitor. *Venture Capital: An International Journal of Entrepreneurial Finance*, 5(2), 101–116.
- Bygrave, W.D. & Timmons, J. (1999). Venture capital: Predictions and outcomes. Venture capital at the crossroads and realizing investment value revisited. In M. Wright & K. Robbie (Eds.), *Management buy-outs and venture capital: Into the next millennium* (pp. 38–56). Northampton, MA: Edward Elgar Publishing.
- Carree, M. & Thurik, R. (2003). The impact of entrepreneurship on economic growth. In D.B. Audretsch & Z.J. Acs (Eds.), *Handbook of entrepreneurship research* (pp. 437–471). Boston: Kluwer Academic Publishers.
- Carree, M., van Stel, A., Thurik, R., & Wennekers, S. (2002). Economic development and business ownership: An analysis using data of 23 OECD countries in the period 1976–1996. *Small Business Economics*, 19, 271–290.
- Casson, M. (1990). *Enterprise and competitiveness*. New York: Oxford University Press.
- CIA World Factbook. (2007). *GDP rank order*. Available at <https://www.cia.gov/library/publications/the-world-factbook/rankorder/2001rank.html>, accessed 15 January 2008.
- De Backer, K. & Sleuwaegen, L. (2003). Does foreign direct investment crowd out domestic entrepreneurship? *Review of Industrial Organization*, 22, 67–84.
- Dodd, S.D. (2002). Metaphors and meaning: A grounded cultural model of US entrepreneurship. *Journal of Business Venturing*, 17(5), 519–536.
- European Commission. (2002). *Benchmarking business angels*. Brussels: The Commission of European Communities.
- Fligstein, N. & Sweet, A.S. (2002). Constructing politics and markets: An institutional account of European integration. *American Journal of Sociology*, 107, 1206–1243.

- Goldstein, H. (1987). *Multi-level models in educational and social research*. London: Griffin.
- Gompers, P. (2005). Venture capital. In B.E. Eckbo (Ed.), *Handbook of corporate finance* (pp. 502–524). Amsterdam: Elsevier/North Holland
- Hayek, F.A. (1945). The use of knowledge in society. *American Economic Review*, 35, 519–530.
- Henderson, J. (2002). Building the rural economy with high-growth entrepreneurs. *Federal Reserve Bank of Kansas City Economic Review*, 87(3), 45–70.
- Higgins, M.C. & Gulati, R. (2006). Stacking the deck: The effects of top management background on investor decisions. *Strategic Management Journal*, 27(1), 1–25.
- Hill, C. (2002). *International business: Competing in the global market place*. New York: McGraw Hill.
- Hwang, H. & Powell, W.W. (2005). Institutions and entrepreneurship. In S.A. Alvarez, R. Agarwal, & O. Sorenson (Eds.), *Handbook of entrepreneurship research: Disciplinary perspectives* (pp. 179–210). New York: Springer.
- Hyder, A.S. & Abraha, D. (2006). Strategic alliances in the Baltic States: A case of Swedish firms. *Competitiveness Review*, 16(3), 173–196.
- Hymer, S.H. (1960). *International operations of national firms, a study of direct foreign investments*. Boston: Massachusetts Institute of Technology.
- International Monetary Fund, Statistics Department. (2004). *Direction of trade statistics yearbook*. Washington, DC: Author .
- Jeng, L. & Wells, P. (2000). The determinants of venture capital funding: Evidence across countries. *Journal of Corporate Finance*, 6, 241–289.
- Jensen, N. (2003). Democratic governance and multinational corporations: Political regimes and inflows of foreign direct investment. *International Organization*, 57(3), 587–616.
- Johanson, J. & Vahlne, J.E. (1977). The internationalization process of the firm—A model of knowledge development and increasing foreign market commitments. *Journal of International Business Studies*, 8(1), 23–32.
- Johnson, R. & Wichern, D. (2002). *Applied multivariate statistical analysis*. Upper Saddle River, NJ: Prentice Hall.
- Jones, K. (2006). Globalization and entrepreneurship. In M. Minniti (Ed.), *Entrepreneurship and entrepreneurial behavior* (Vol. I, pp. 200–222). New York: Praeger.
- Kirzner, I.M. (1973). *Competition and entrepreneurship*. Chicago: University of Chicago Press.
- Kirzner, I.M. (1979). *Perception, opportunity, and profit: Studies in the theory of entrepreneurship*. Chicago: University of Chicago Press.
- Kreft, S.F. & Sobel, R.S. (2005). Public policy, entrepreneurship, and economic freedom. *Cato Journal*, 25(3), 595–616.
- Krugman, P.R. & Obstfeld, M. (2002). *International economics: Theory and policy*. Boston: Addison Wesley.
- Krull, J.L. & MacKinnon, D.P. (2001). Multilevel modeling of individual and group level mediated effects. *Multivariate behavioral research*, 36(2), 249–277.
- Leblond, P. (2004). Completing the maastricht contract: Institutional handicraft and the transition to European monetary union. *Journal of Common Market Studies*, 42(3), 553–572.

- Leonidou, L.C. & Katsikeas, C.S. (1996). The export development process: An integrative review of empirical models. *Journal of International Business Studies*, 27(3), 517–551.
- Levy, B. (2006). Emerging countries, regionalization, and world trade. *Global Economy Journal*, 6(4), 1–29.
- Manalova, T.S., Eunni, R.V., & Gyoshev, B.S. (2008). Institutional environments for entrepreneurship: Evidence from emerging economies in Eastern Europe. *Entrepreneurship Theory and Practice*, 32(1), 203–218.
- Manigart, S. (1994). The founding rates of venture capital firms in three European countries (1970–1990). *Journal of Business Venturing*, 9(6), 525–541.
- Mason, C.M. & Harrison, R.T. (2000). The size of the informal venture capital market in the United Kingdom. *Small Business Economics*, 15(2), 137–148.
- Maula, M. & Mäkelä, M. (2003). Cross-border venture capital? In A. Hyytinen & M. Pajarinen (Eds.), *Financial systems and firm performance: Theoretical and empirical perspectives* (pp. 269–291). Helsinki: Taloustieto.
- Meggison, W. (2004). Toward a global model of venture capital? *Accenture Journal of Applied Corporate Finance*, 16(1), 7–26.
- Meyer, J. & Shao, J. (1995). International venture capital portfolio diversification and portfolio costs. *Multinational Business Review*, 3, 53–58.
- Minniti, M. (1999). Entrepreneurial activity and economic growth. *Global Business and Economics Review*, 1(1), 31–42.
- Mitchell, R.K. (2003). A transaction cognition theory of global entrepreneurship. In J.A. Katz & D.A. Shepherd (Eds.), *Advances in entrepreneurship, firm emergence and growth: Cognitive approaches to entrepreneurship research* (pp. 183–231). Greenwich, CT: JAI Press.
- Mitchell, R.K., Smith, B., Seawright, K.W., & Morse, E.A. (2000). Cross-cultural cognitions and the venture creation decision. *Academy of Management Journal*, 43(5), 974–993.
- Mitchell, R.K., Smith, J.B., Morse, E.A., Seawright, K.W., Peredo, A.M., & McKenzie, B. (2002). Are entrepreneurial cognitions universal? Assessing entrepreneurial cognitions across cultures. *Entrepreneurship Theory and Practice*, 26(4), 9–32.
- Nelson, R.R. (1982). *Government and technical progress: A cross-industry analysis*. New York: Pergamon.
- Nickell, S.J. (1996). Competition and corporate performance. *Journal of Political Economy*, 104, 724–746.
- Reynolds, P.D., Bosma, N., Autio, E., Hunt, S., De Bono, N., Servais, I., et al. (2005). Global entrepreneurship monitor: Data collection design and implementation, 1998–2003. *Small Business Economics*, 24, 205–231.
- Reynolds, P.D., Bygrave, W.D., Autio, E., Cox, L.W., & Hay, M. (2002). *Global entrepreneurship monitor: 2002 executive report*. Kansas City, MO: Kauffman Center for Entrepreneurial Leadership.
- Reynolds, P.D., Hay, M., & Camp, M. (1999). *Global entrepreneurship monitor: 1999 executive report*. Kansas City, MO: Kauffman Center for Entrepreneurial Leadership.
- Rose, A.K. (2000). One money, one market: The effect of common currency on trade. *Economic Policy*, 30, 8–45.
- Rose, A.K. & Van Wincoop, E. (2001). National money as a barrier to international trade: The real case for currency union. *American Economic Review*, 91(2), 386–390.
- Salvatore, D. (2002). The Euro, the European Central Bank, and the International Monetary System. *Annals of the American Academy of Political and Social Science*, 579, 153–167.

- Scharpf, F. (1996). Negative and positive integration in the political economy of European welfare states. In G. Marks, F.W. Scharpf, P.C. Schmitter, & W. Streeck (Eds.). *Governance in the European Union* (pp. 15–40). London: Sage.
- Schumpeter, J.A. (1934). *The theory of economic development*. Cambridge, MA: Harvard University Press.
- Shane, S. (1992). Why do some countries invent more than others? *Journal of Business Venturing*, 7, 29–46.
- Thomas, A. & Mueller, S. (2000). A case of comparative entrepreneurship: Assessing the relevance of culture. *Journal of International Business Studies*, 31(2), 287–301.
- van Stel, A., Carree, M., & Thurik, R. (2005). The effect of entrepreneurial activity on national economic growth. *Small Business Economics*, 24, 311–321.
- World Bank. (2005). *Global economic prospects: Trade, regionalism and development*. Washington, DC: The International Bank for Reconstruction and Development/The World Bank.
- Wright, M., Pruthi, S., & Lockett, A. (2005). International venture capital research: From cross-country comparisons to crossing borders. *International Journal of Management Reviews*, 7(3), 135–165.
- Wright, M., Robbie, K., Romanet, Y., Thompson, S., Joachimsson, R., Bruining, H., et al. (1993). Harvesting and the longevity of management buyouts and buy-ins: A four country study. *Entrepreneurship Theory and Practice*, 18(2), 90–109.
- Yiu, D. & Makino, S. (2002). The choice between joint venture and wholly owned subsidiary: An institutional perspective. *Organization Science*, 13(6), 667–683.
- Zacharakis, A.L., Bygrave, W.D., & Shepherd, D.A. (2000). *Global entrepreneurship monitor: National entrepreneurship assessment: United States of America*. Kansas City, MO: Kauffman Center for Entrepreneurial Leadership.
- Zacharakis, A.L. & Shepherd, D.A. (2001). The nature of information and overconfidence on venture capitalists' decision making. *Journal of Business Venturing*, 16(4), 311–332.

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